

Research Assignment

Title of the Research Assignment

Optional subtitle

Forename Surname

Matriculation number: 12345678

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1 Introduction

Introduce the project objective and its context.

2 Methods

Describe the approach and implementation.

3 Results

Present and discuss the results.

4 Conclusion

Summarize the main outcome and remaining limitations.

References

- [1] Kaiming He et al. "Deep residual learning for image recognition". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016, pp. 770–778.

A LaTeX tips and examples

A.1 General

- Use present tense in general: “For evaluation, we use the F_1 score ...” “The advantage of the method is its lower memory use ...” and “We find that performance improves with more training data.” *etc.*
- Use past tense for completed experiments: “We trained the model for 100 epochs.”, “We collected 1000 samples from the dataset.” *etc.*
- Cite entries from `references.bib` by their key, *e.g.* `\cite{he2016deep}` produces [1]
- Use the macros from `definitions.tex` for abbreviations such as `\eg`, `\ie`, and `\etc`, and for the number sets `\R`, `\Z`, `\N`, `\Q`, and `\C`. For example, `$x \in \R^n$` produces $x \in \mathbb{R}^n$
- Add a `\label` to sections, equations, figures, and tables and refer to them with `\ref`, *e.g.* Equation 1. Do not type reference numbers manually
- In general, refer to each figure and table in the text before it appears

A.2 Equations

Use `$...$` for inline math, *e.g.* `$F_1$` produces F_1 . Use the `equation` environment for numbered equations:

$$F_1 = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}, \quad (1)$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives.

A.3 Figures

In general, use wide figures and place them at the top or bottom of a page. *E.g.* Figure 1 uses the full text width, while Figure 2 uses subfigures. Put captions below figures and briefly explain all panels, symbols, units, *etc.*

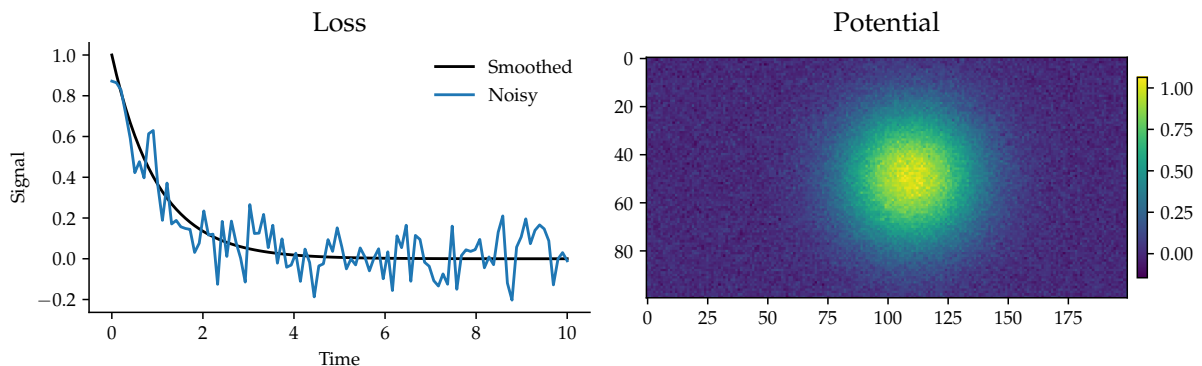


Figure 1: **Example figure.** Line plot showing a noisy signal and its smoothed trend (left). Heat map showing a spatial intensity distribution (right).

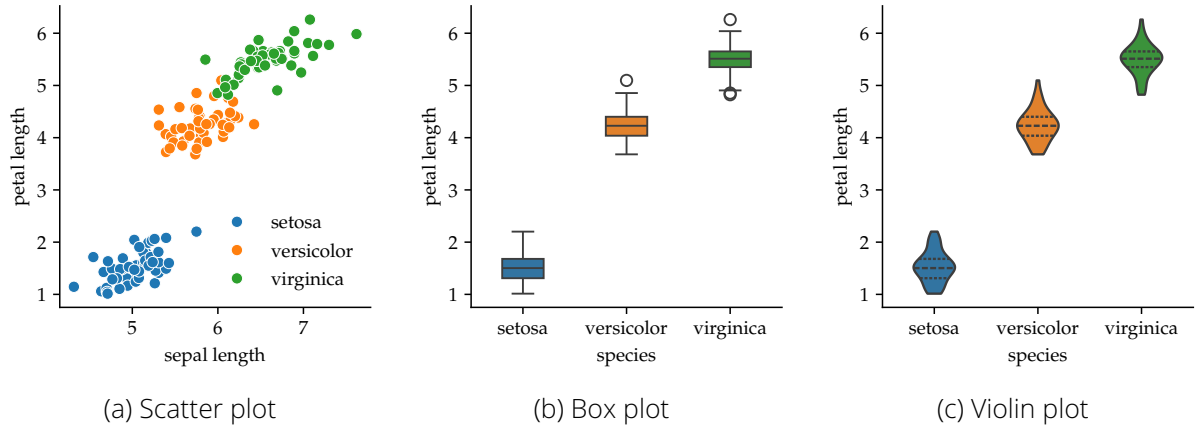


Figure 2: **Alternative visualizations of the same data.** Each panel shows the same synthetic Iris-style dataset.

A.4 Tables

Put table captions above the table. Use booktabs, omit vertical lines, keep decimal places consistent, and bold the best value in each column, as in Table 1.

Table 1: **Example quantitative comparison.** We show accuracy, precision, and recall (higher is better) as well as error (lower is better) for three methods.

Method	Accuracy $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	Error $\downarrow$
Method A	0.81	0.80	0.78	0.23
Method B	0.85	0.86	0.84	0.19
Method C	0.89	0.84	0.82	0.15